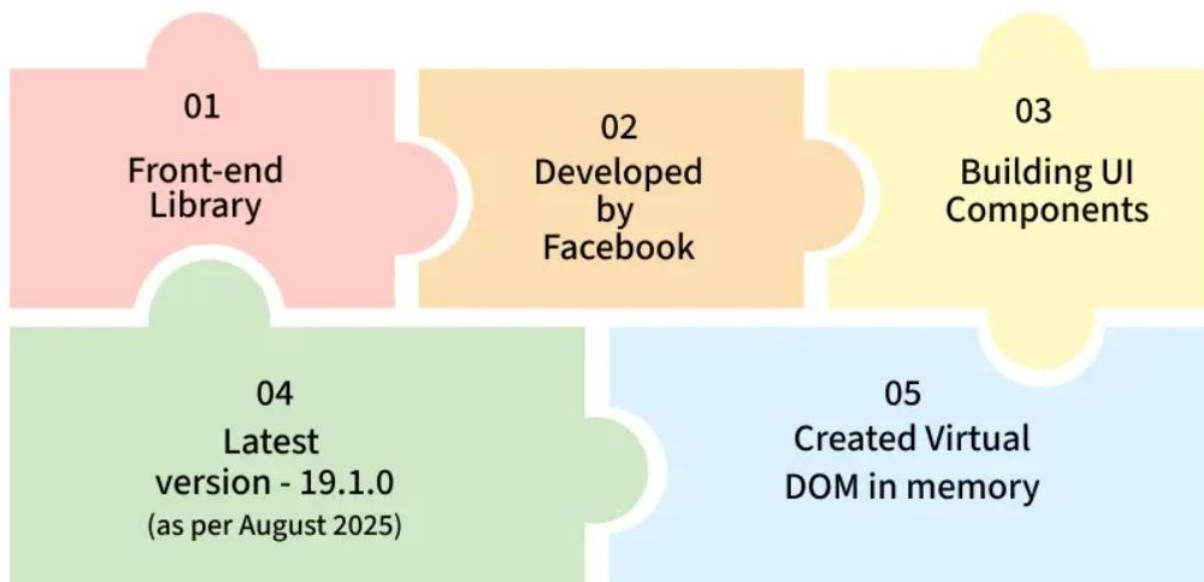


# Introduction to Basics of React



ReactJS is a component-based JavaScript library used to build dynamic and interactive user interfaces. It simplifies the creation of single-page applications (SPAs) with a focus on performance and maintainability.



## Why React Was Created

- Earlier, websites were mostly **static** (not changing much).
- If developers wanted to change something, they had to **edit many files**.
- React made websites **dynamic**, meaning they can **update automatically when data changes**.

Example:

- Online shopping cart updating items.
- Live news updates on a website.

## Virtual DOM

React keeps a **smart copy of the webpage** called the **Virtual DOM**.

- It checks what changed and updates only that part.

### Simple Real-Life Example

Think of React like a **chef's kitchen**:

- Ingredients = **Data**
- Recipe = **Code**
- If ingredients change, the dish updates quickly without changing the whole kitchen.

## Steps for creating a React app

### Step 1: Prerequisites

Before creating a React app, make sure these are installed:

1. Install **Node.js** from <https://nodejs.org>
2. **npm (Node Package Manager)** will be installed automatically with Node.js.

### Step 2: Create a New React App

Open **Command Prompt / Terminal** and run:

```
npx create-react-app myfirstapp
```

What this does:

- npx runs npm commands without installing globally.
- create-react-app creates the React project automatically.
- A new folder named **myfirstapp** will be created.

### Step 3: Navigate to the Project Folder

- Move inside the project folder:
- `cd myfirstapp`

### Step 4: Start the React App

- Run the following command:
- `npm start`

### Step 5: React Folder Structure

- Main folders and files inside the project:
- myfirstapp/
  - node\_modules/ → Stores installed packages
  - public/ → Public files (index.html)
  - src/
    - App.js → Main component
    - index.js → Entry point of the app
    - App.css → Styling for App.js
  - package.json → Project dependencies

### Step 6: Editing Your First React Component

Open App.js inside the src folder. You will see the following code:

Code

```
function App() {  
  return (  
    <div>  
      <h1>Hello, React!</h1>  
      <p>This is my first React app.</p>  
    </div>  
  );  
}  
  
export default App;
```

#### Explanation:

- App() is a component, which is a reusable block of code.
- It returns JSX, which looks like HTML but can include JavaScript expressions.
- export default App; allows this component to be used in other files.

### Step 7: Stopping the App

To stop the running React app, press **Ctrl + C** in the terminal.

## Applications of React

- Web Development
- Mobile Apps
- Enterprise Applications
- Dashboards and Data Visualizations

**1. What is React?**

- A. A database
- B. A JavaScript library for building user interfaces
- C. A programming language
- D. An operating system

**2. Which company created React?**

- A. Google
- B. Microsoft
- C. Facebook
- D. Amazon

**3. Which command is used to create a new React app?**

- A. npm start
- B. node install react
- C. npx create-react-app myapp
- D. react new app

**4. Which command starts a React application?**

- A. npm start
- B. npm run react
- C. node start
- D. react start

**5. Which folder contains the main React code?**

- A. public
- B. src
- C. node\_modules
- D. package